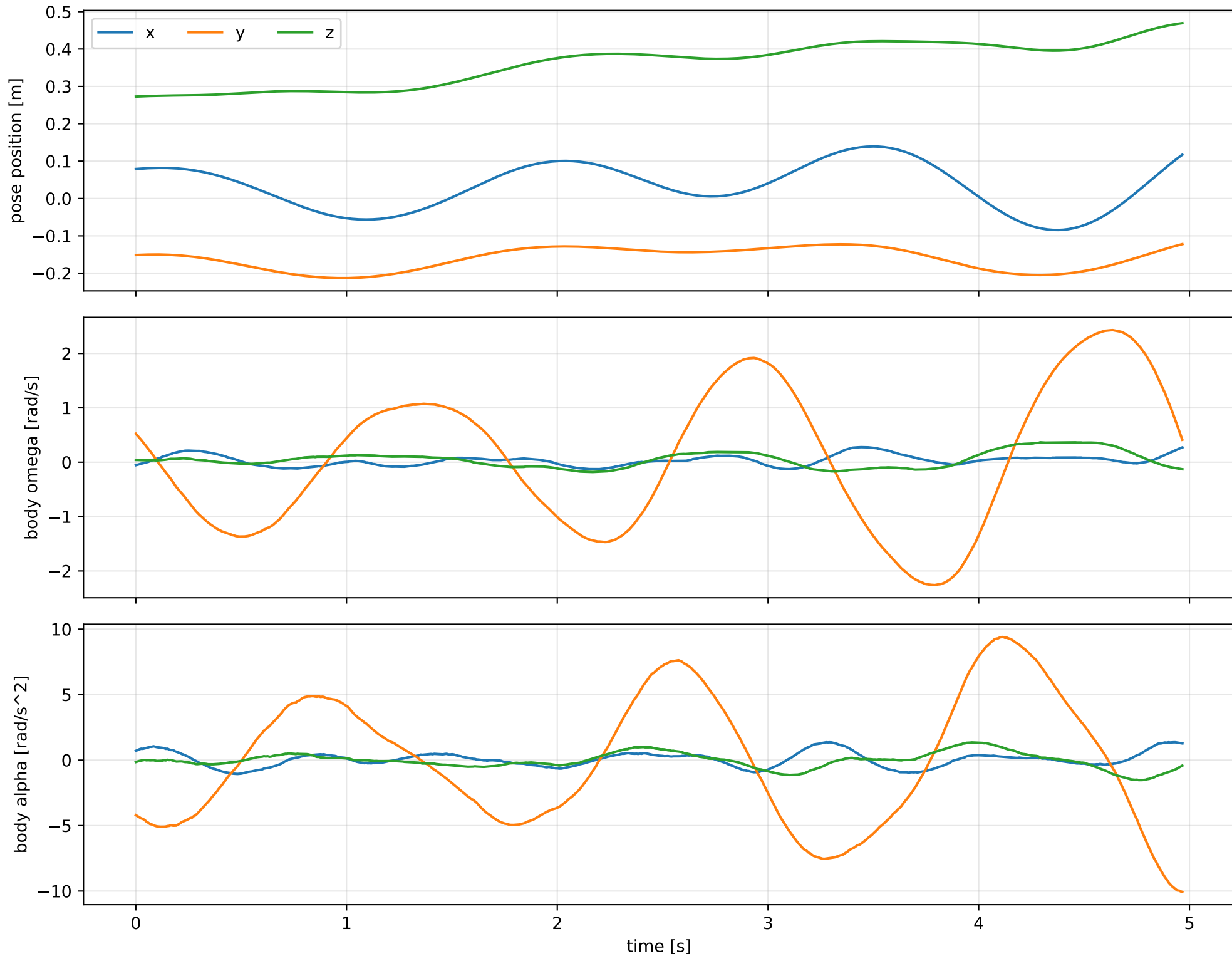
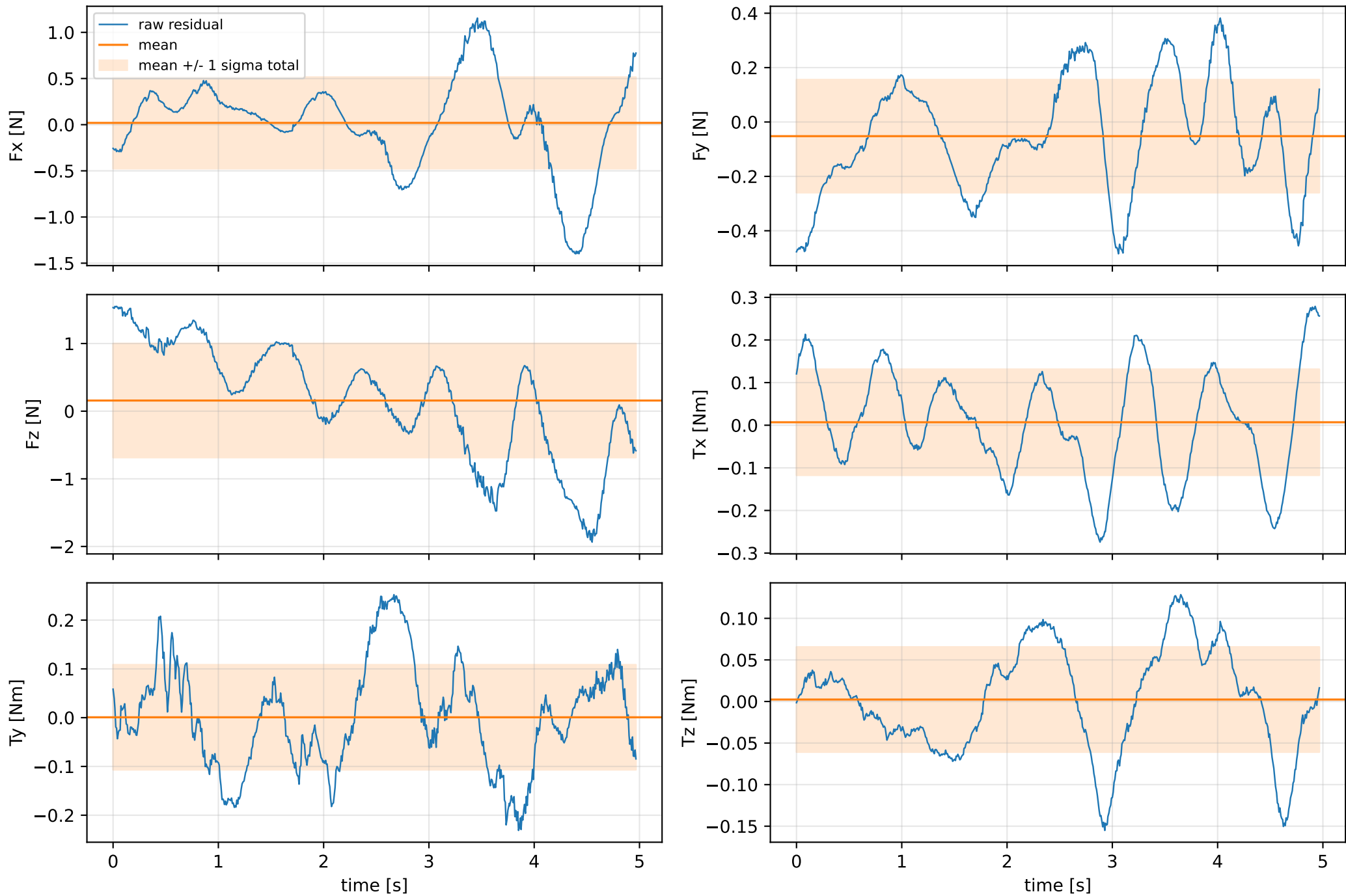


inertia_over_mass_zz_nominal: SG trajectory and derived motion



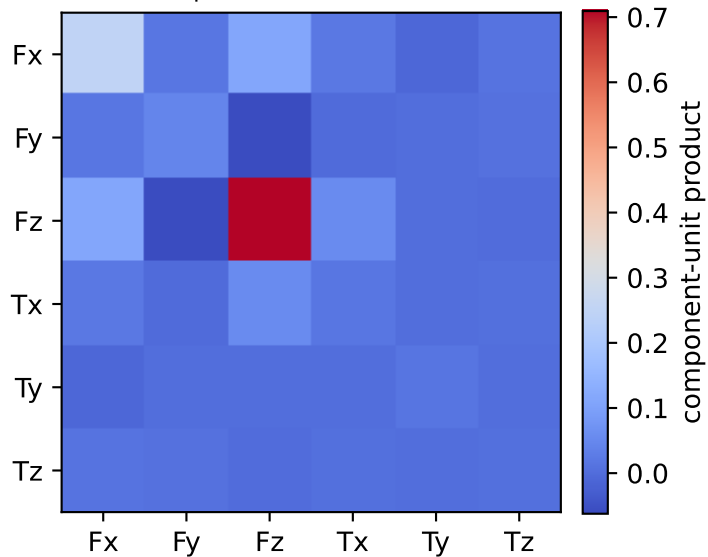
inertia_over_mass_zz_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



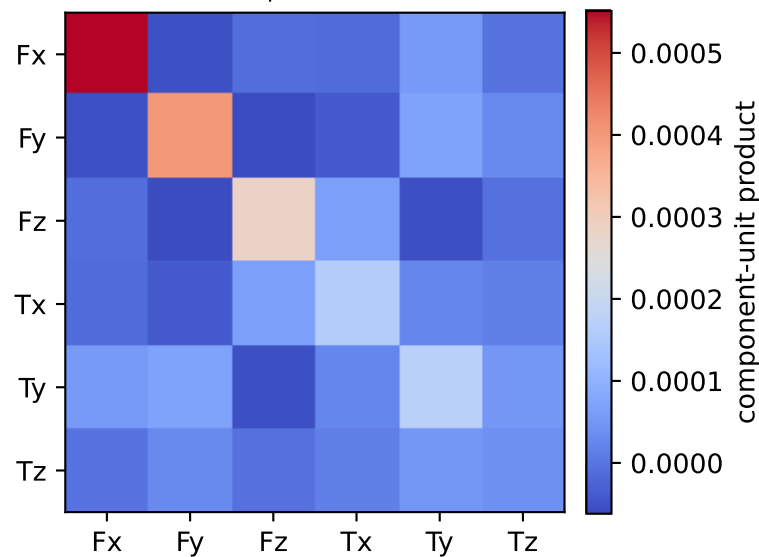
vector RMS about zero: force 1.012 N, torque 0.17612 Nm; component std about mean: force [0.4955 0.2077 0.8422], torque [0.1243 0.1076 0.0632]

inertia_over_mass_zz_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.003 0.01 0.011 0.035 0.225 0.744]

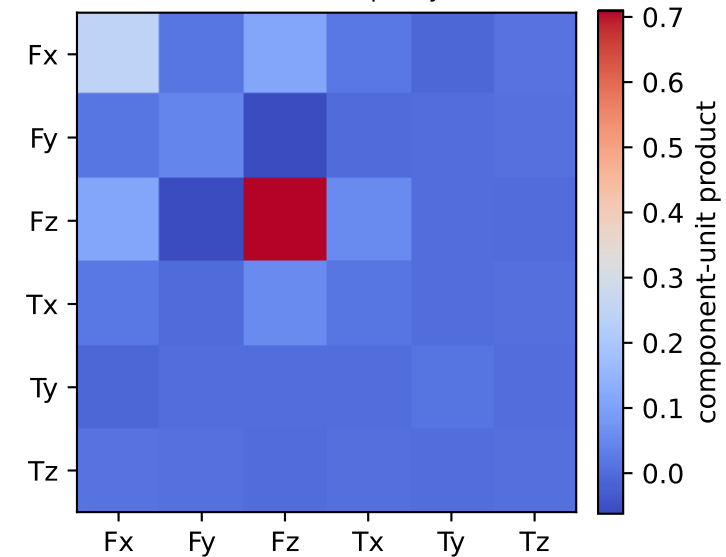
empirical total covariance



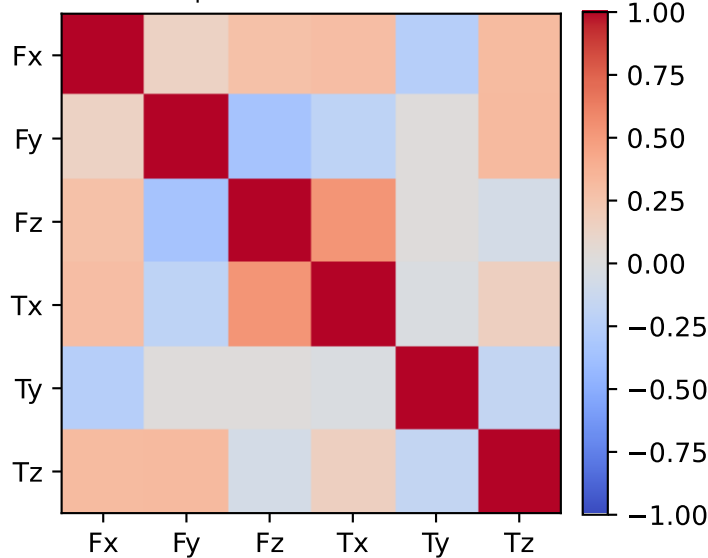
mean full-SG prediction covariance



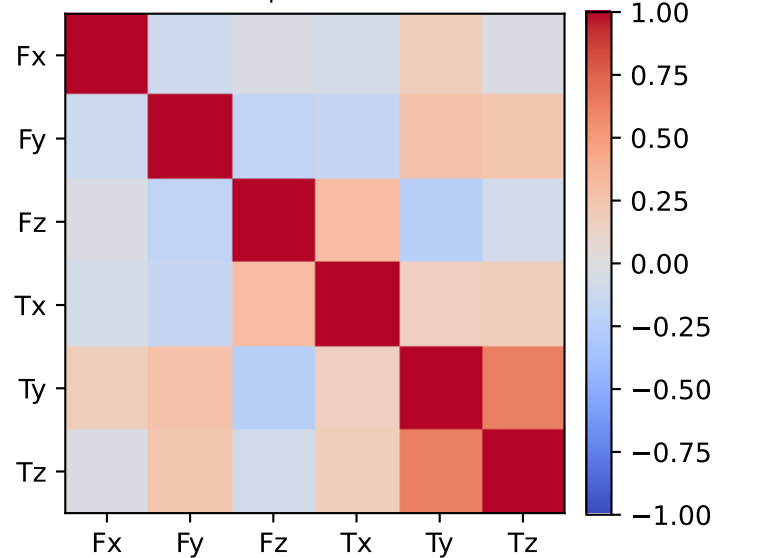
PSD excess model discrepancy covariance



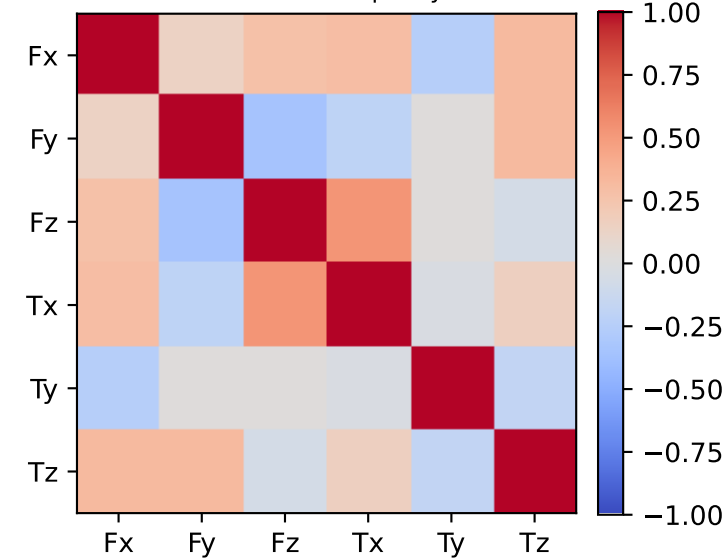
empirical total correlation



mean full-SG prediction correlation

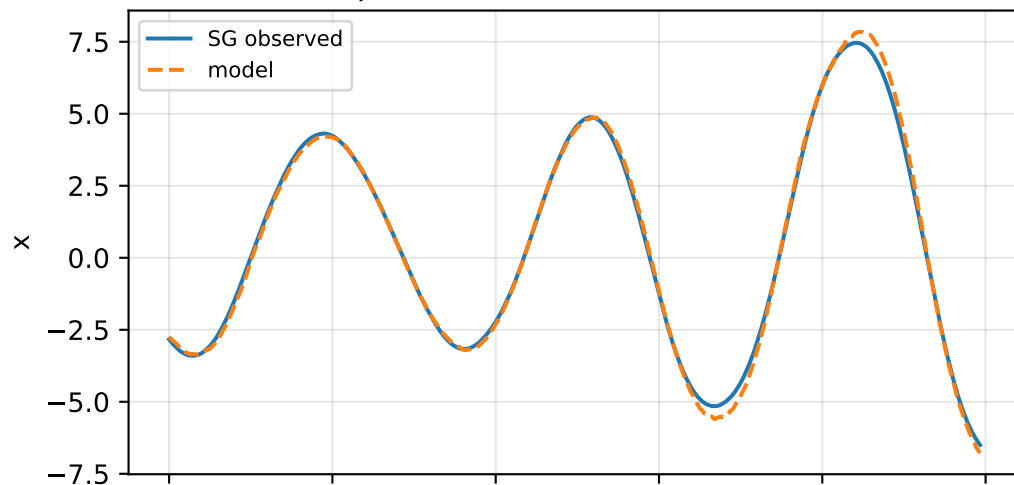


PSD excess model discrepancy correlation

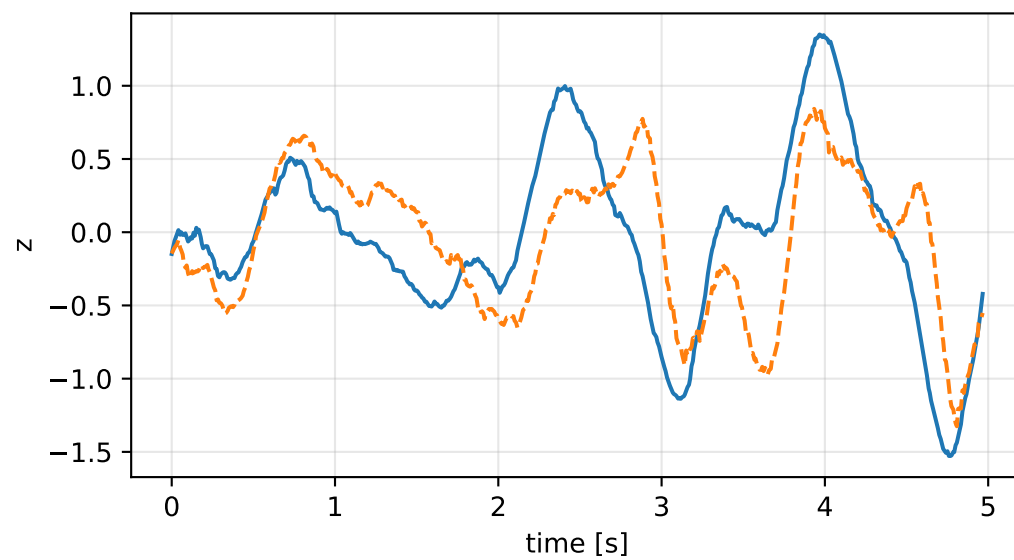
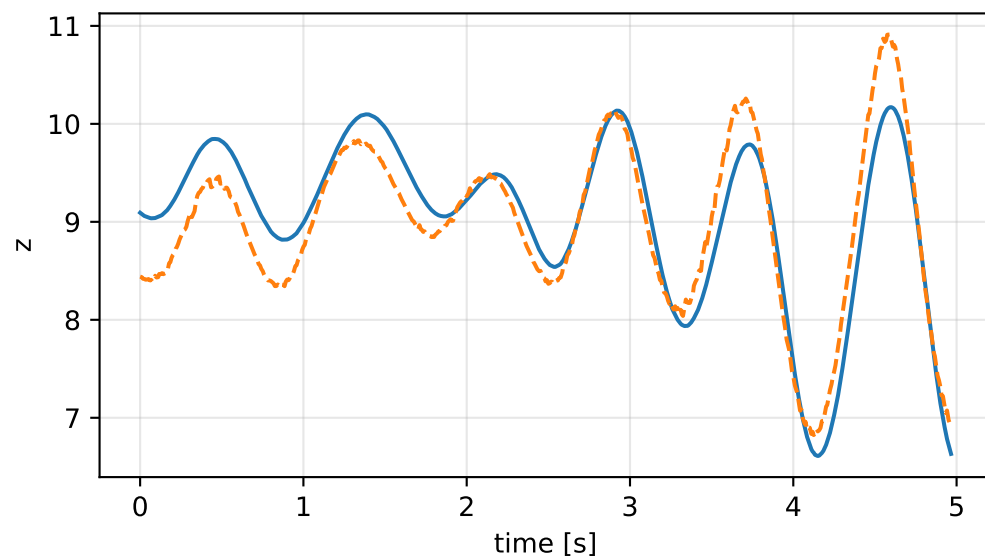
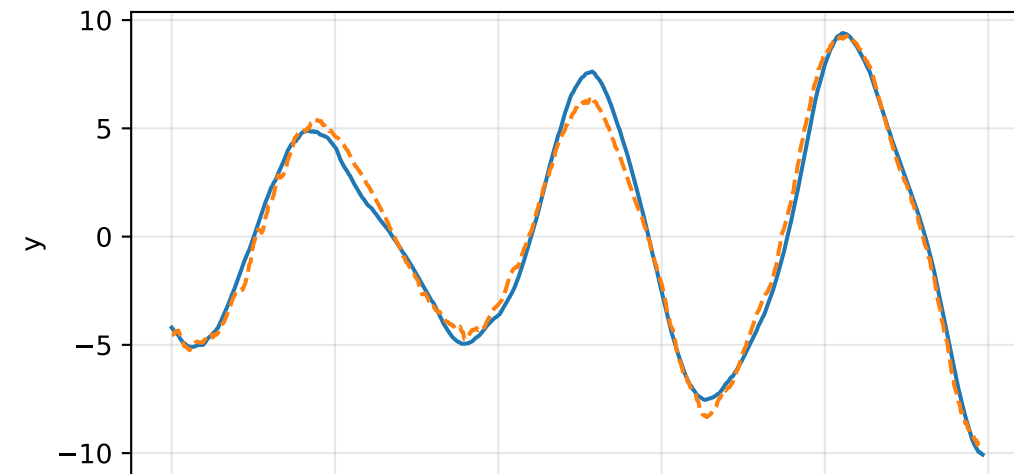
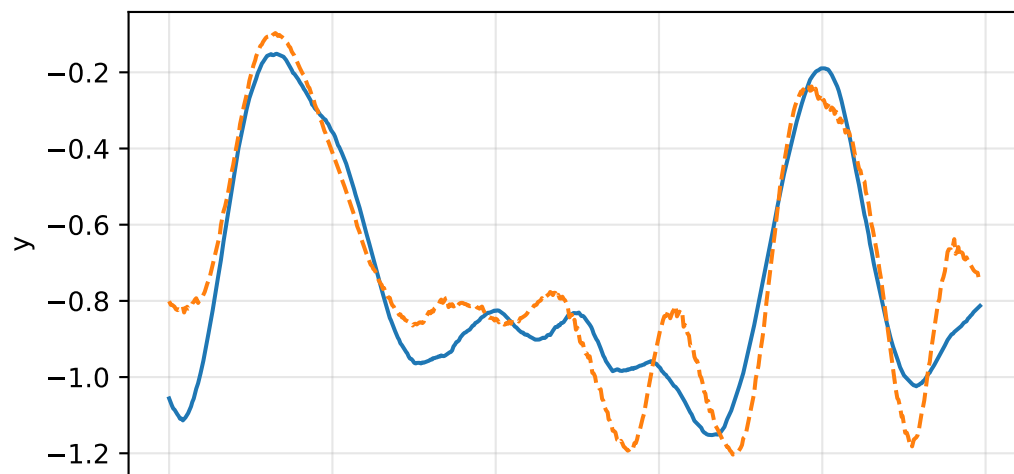
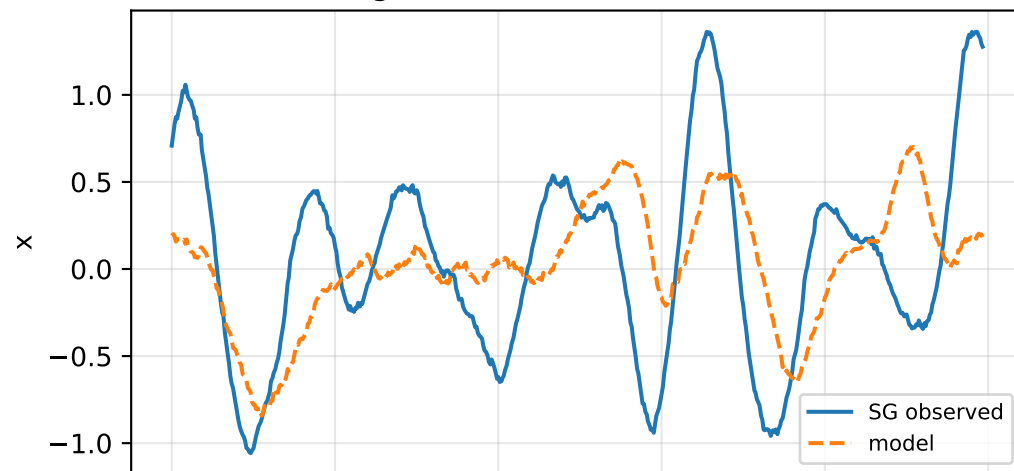


inertia_over_mass_zz_nominal: acceleration residual objective

specific acceleration [m/s^2]

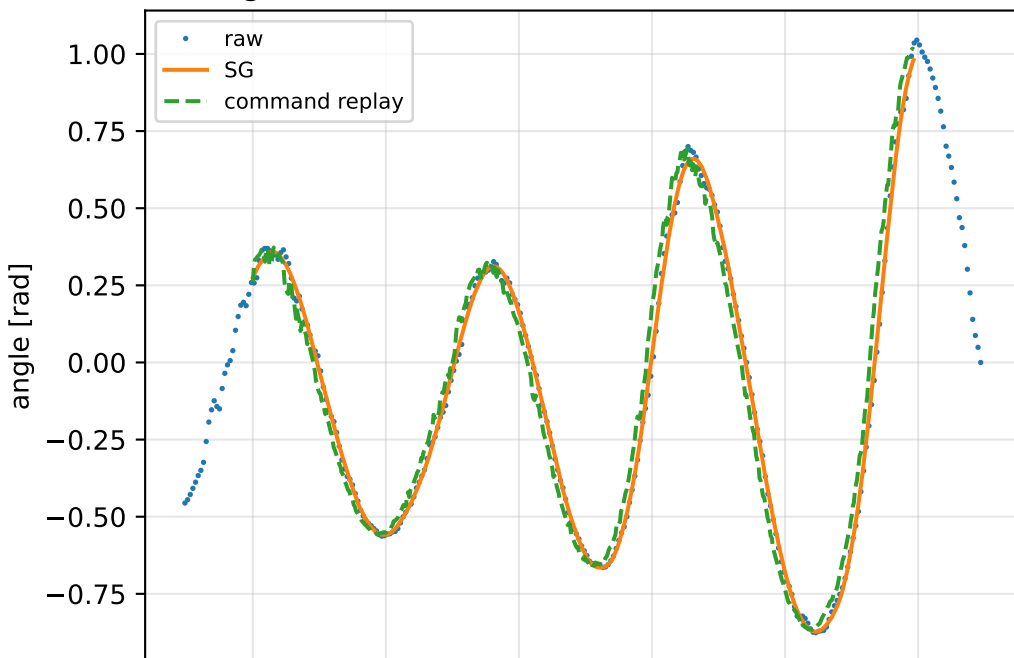


angular acceleration [rad/s^2]

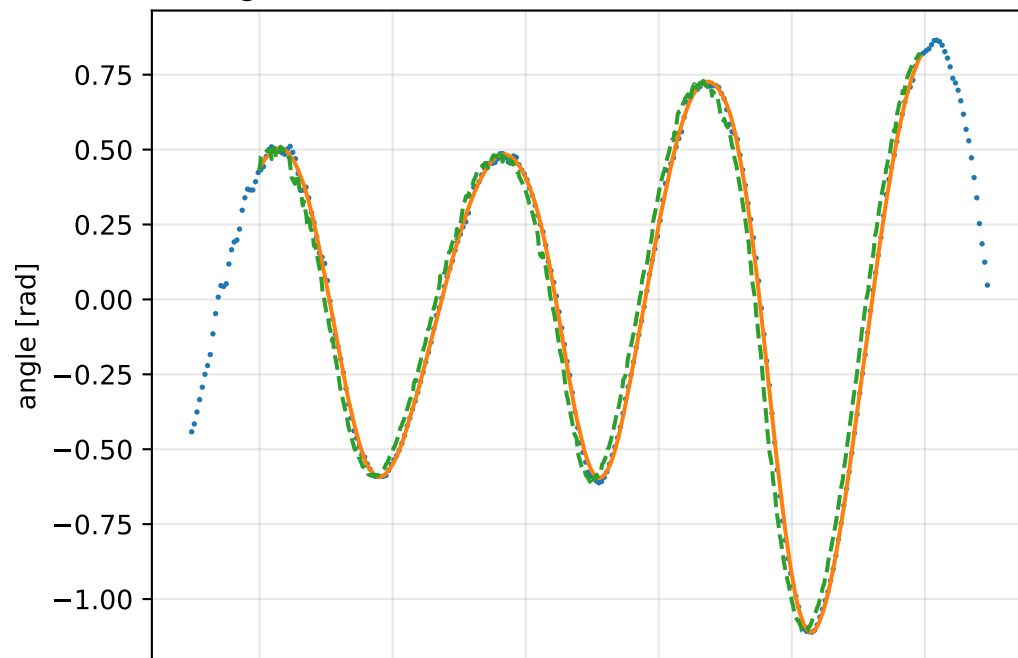


inertia_over_mass_zz_nominal: gimbal measurement audit (objective=measured_sg)

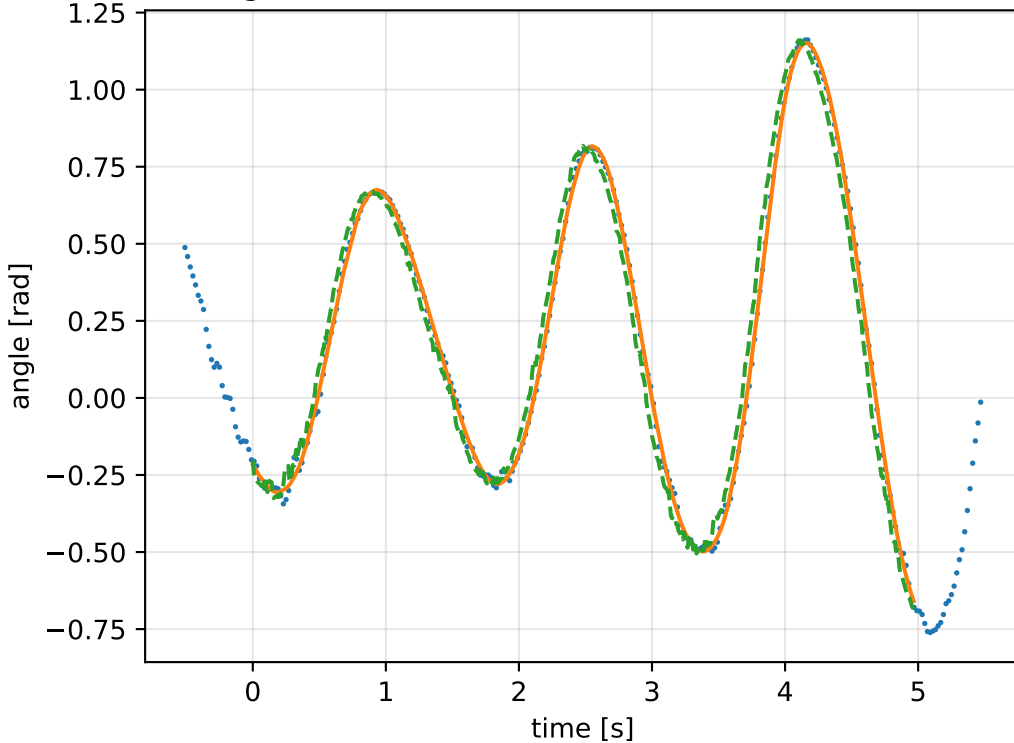
gimbal 1: RMSE=0.0759, max=0.2095 rad



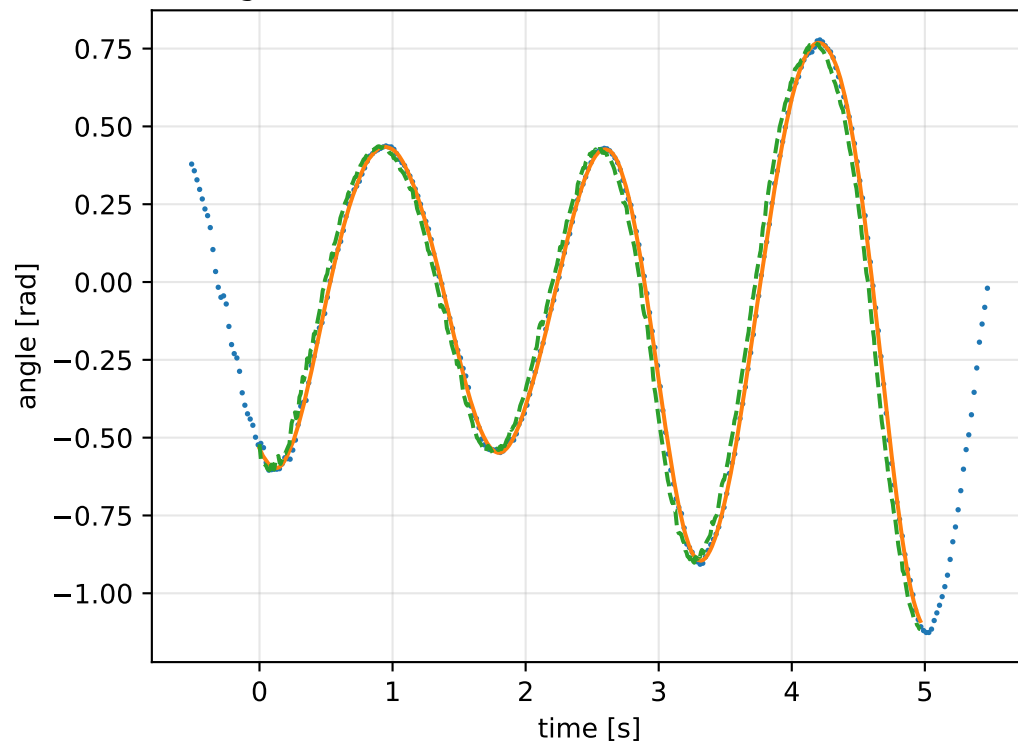
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

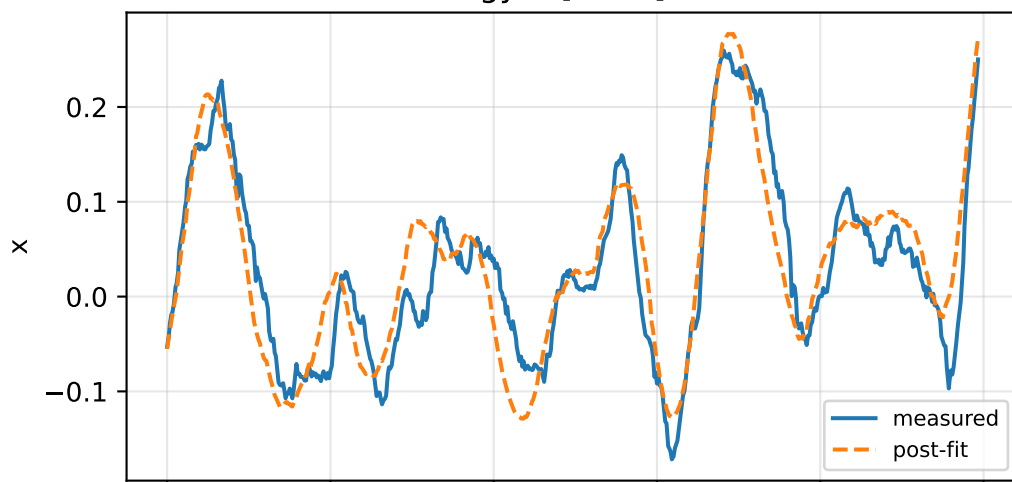


gimbal 4: RMSE=0.07131, max=0.1737 rad

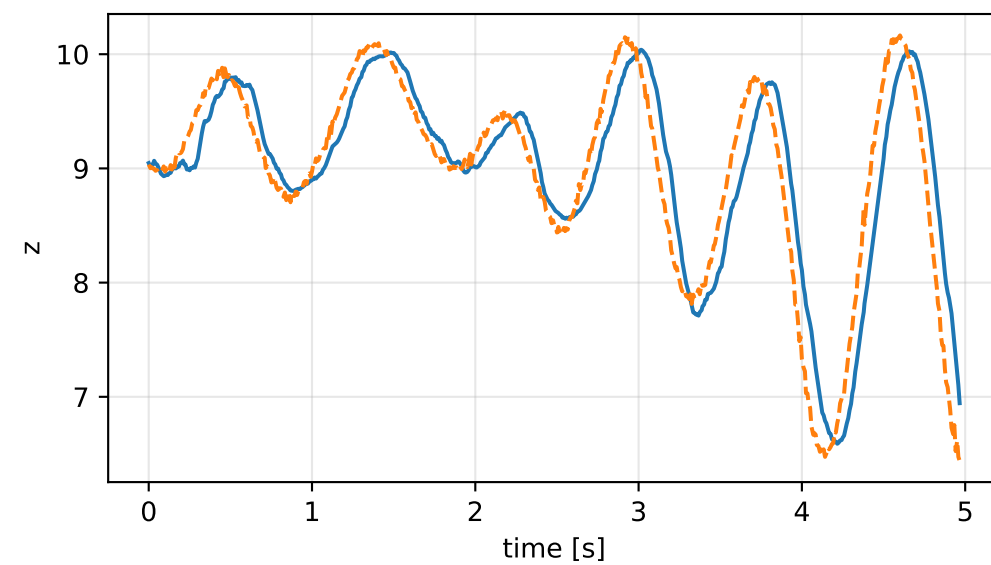
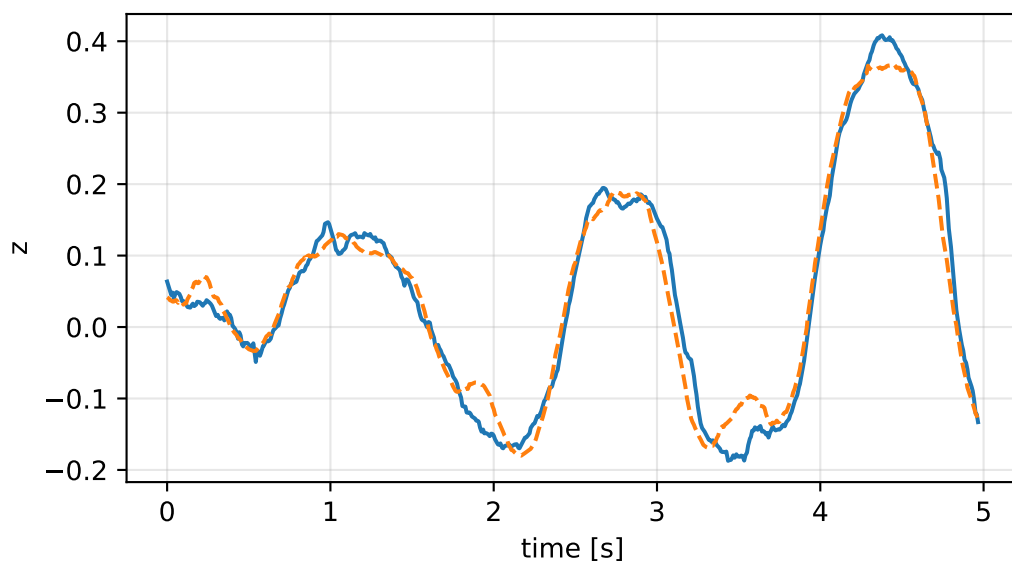
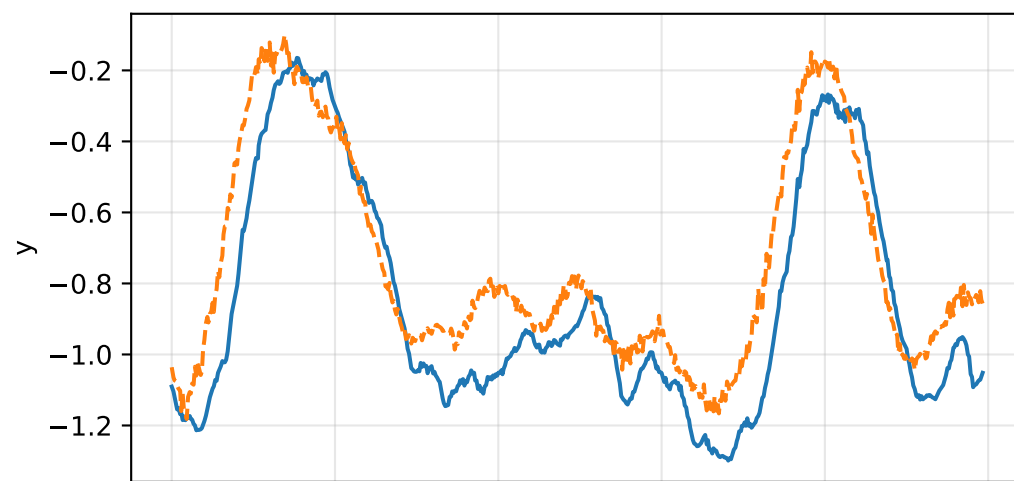
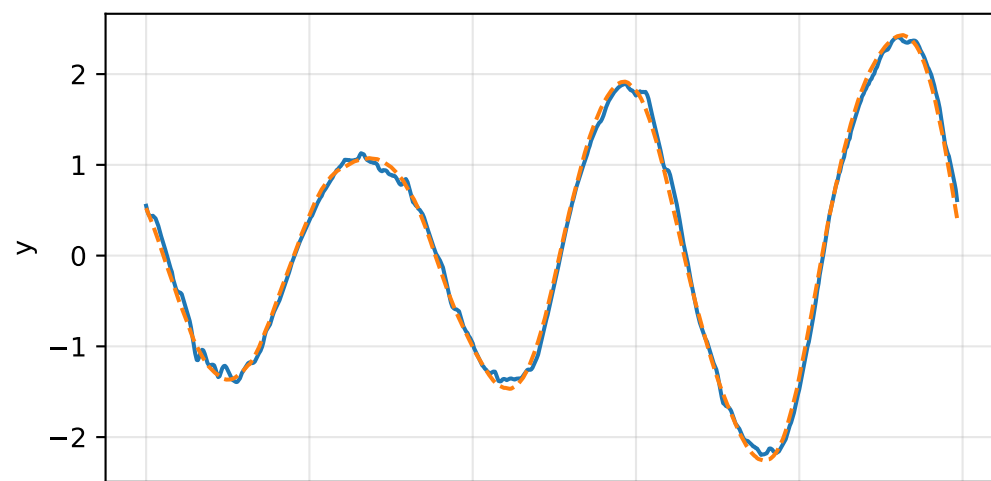
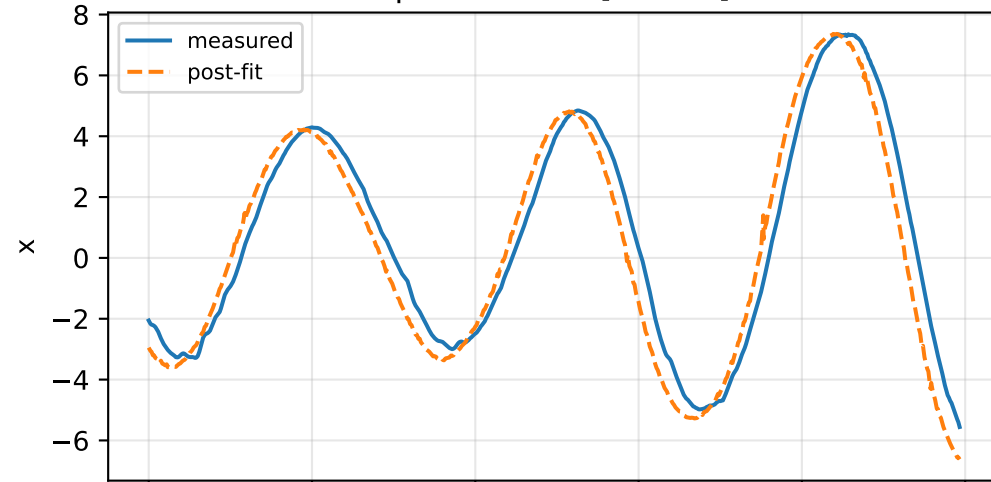


inertia_over_mass_zz_nominal: IMU comparison (diagnostic only)

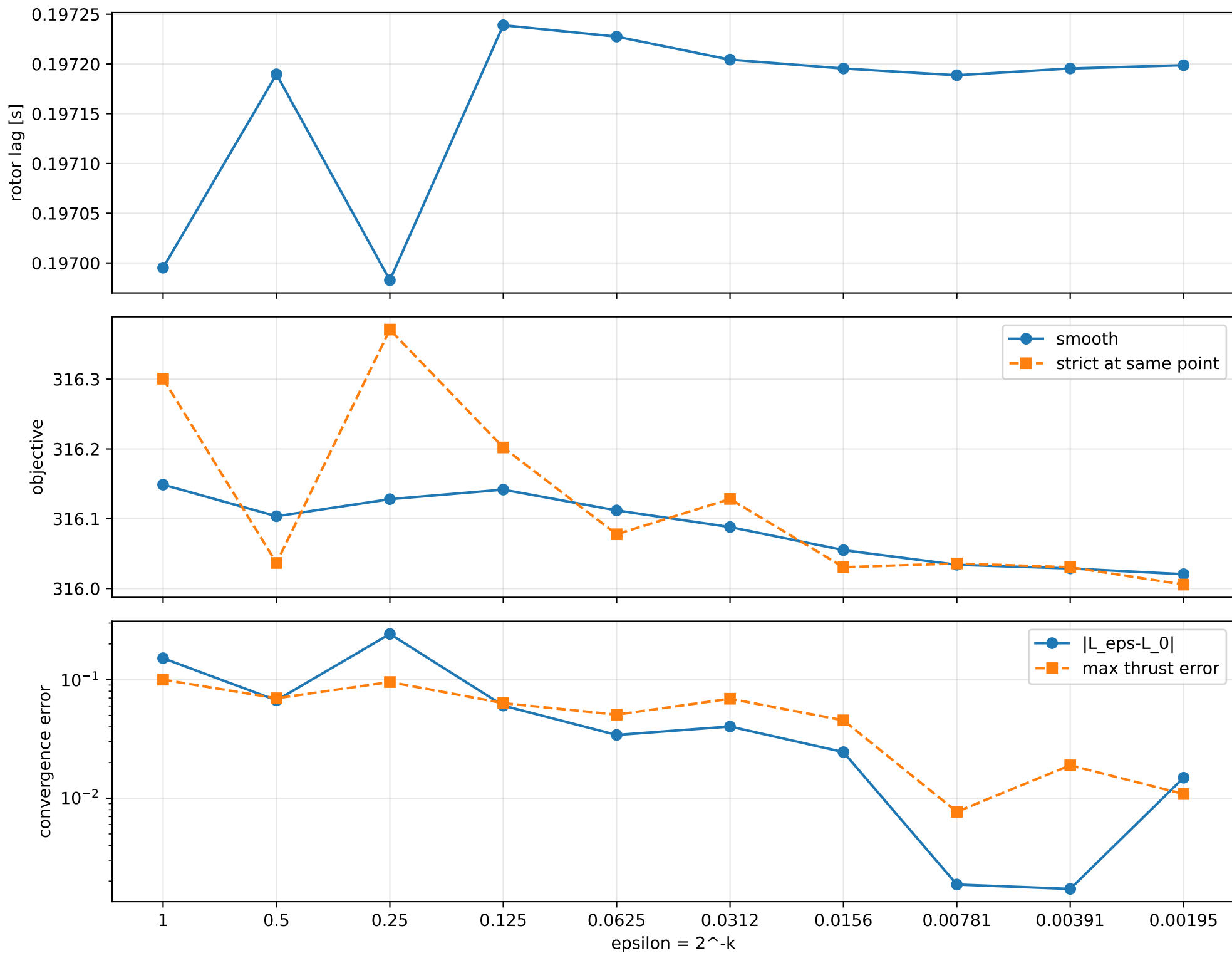
gyro [rad/s]



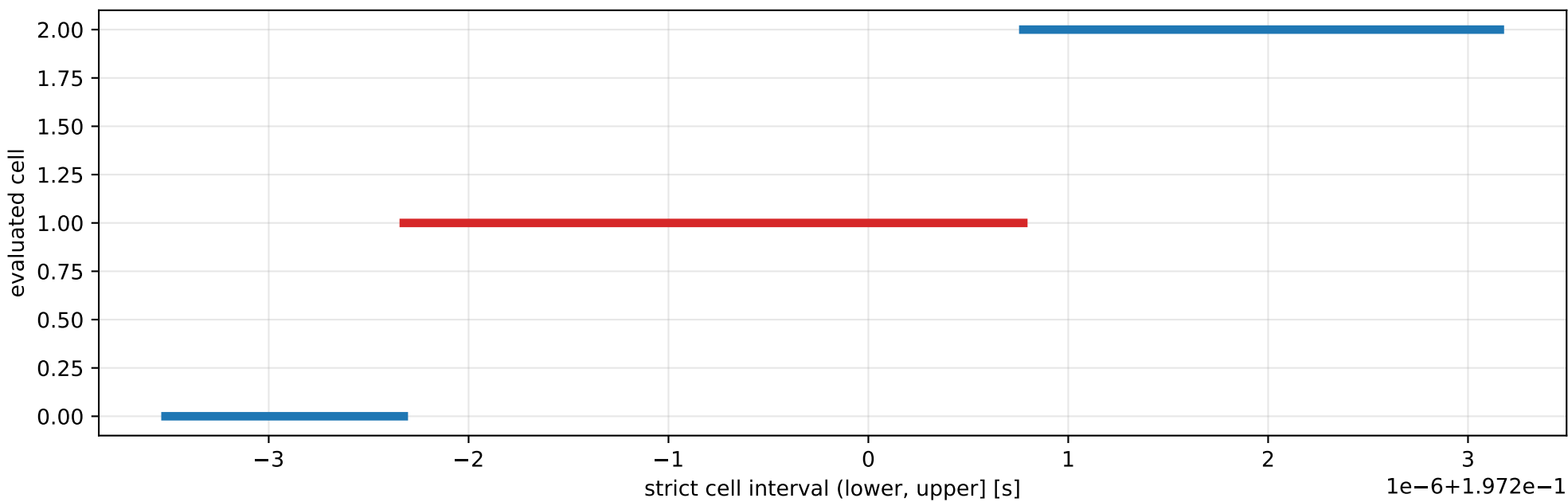
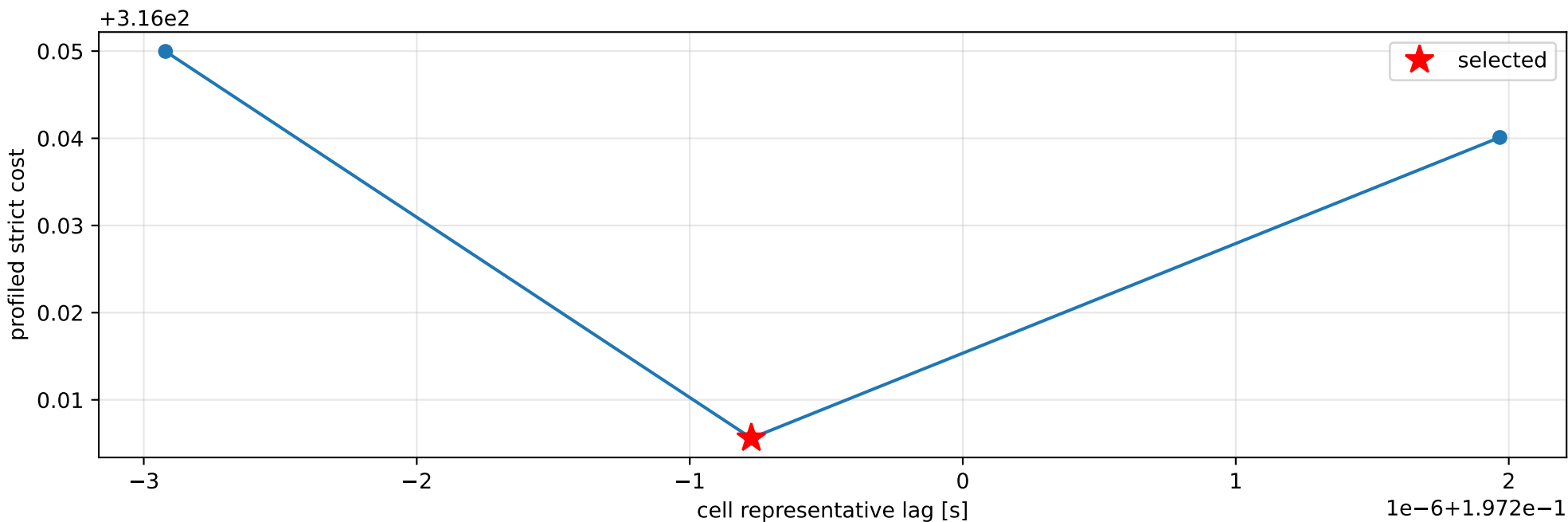
specific force [m/s^2]



inertia_over_mass_zz_nominal: smooth-to-strict continuation



inertia_over_mass_zz_nominal: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197198729 s; data boundary=False

inertia_over_mass_zz_nominal: parameter estimate

Case: inertia_over_mass_zz_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.28021362

inertia [kg m²]:

```
[[ 0.127356 -0.00082927 -0.00055505]
 [-0.00082927 0.09862471 -0.00094668]
 [-0.00055505 -0.00094668 0.07022352]]
```

CoG body [m]: [0.00865194 0.01420613 -0.04343432]

force effectiveness: [0.41742634 0.4921151 0.5012039 0.44638034]

scale-free J/m [m²]:

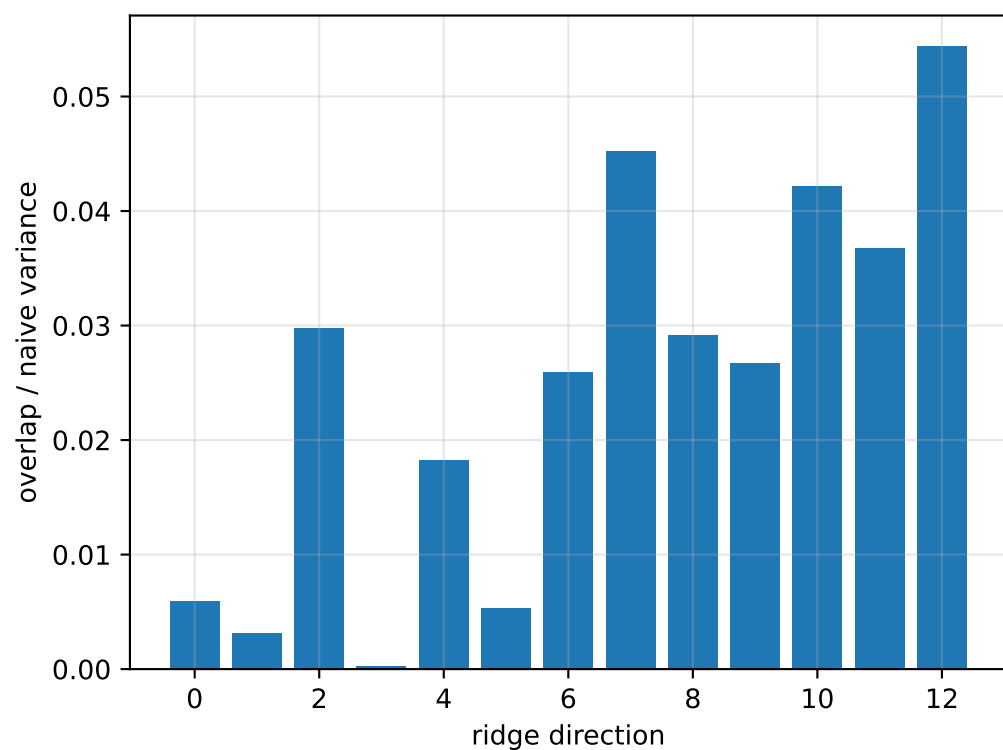
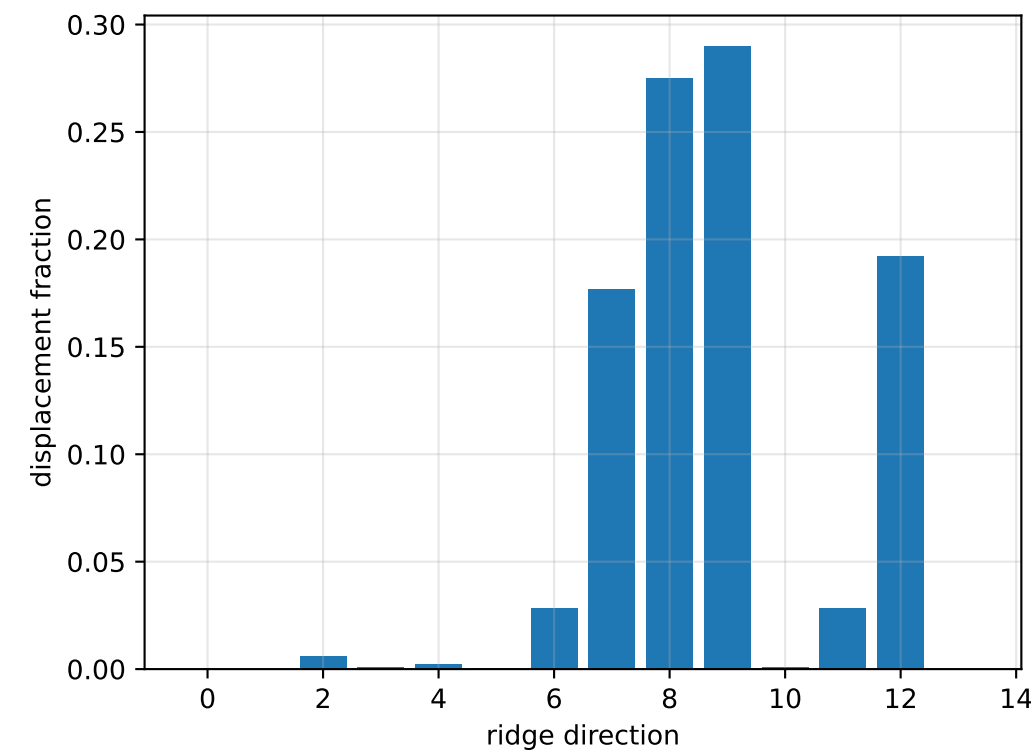
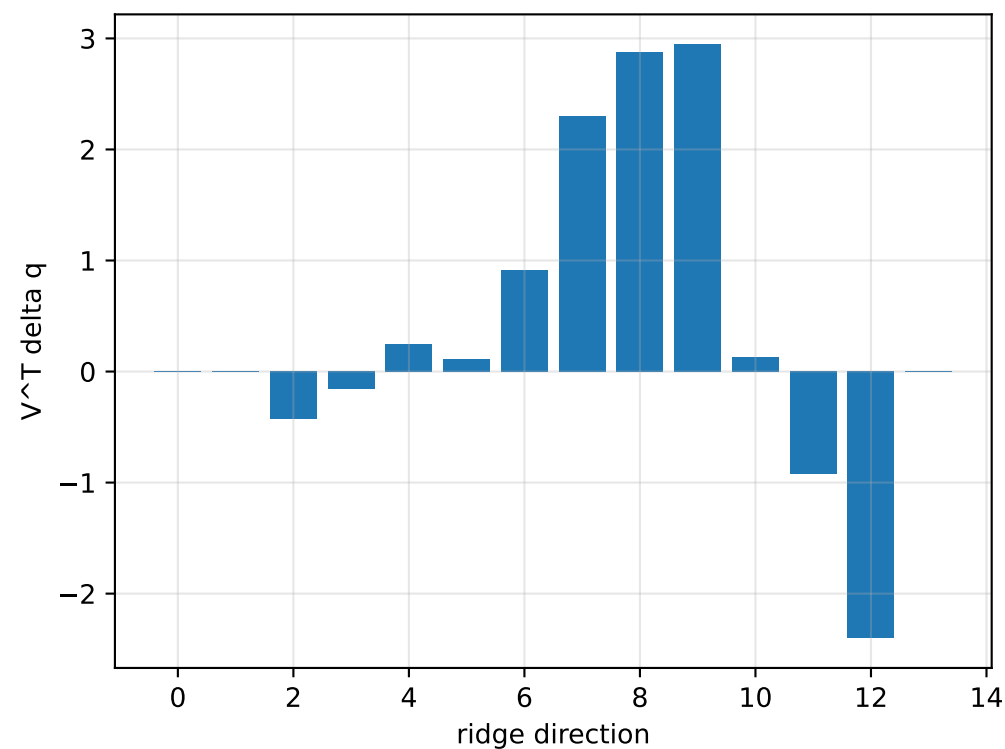
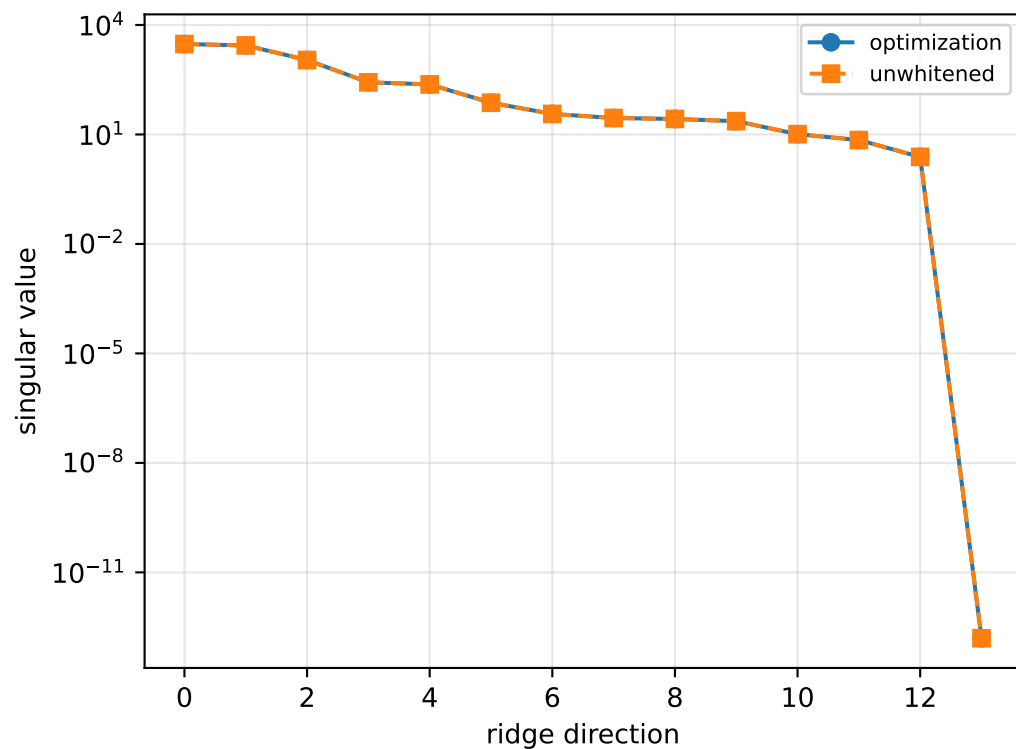
```
[[ 0.09948028 -0.00064776 -0.00043356]
 [-0.00064776 0.0770377 -0.00073947]
 [-0.00043356 -0.00073947 0.05485297]]
```

scale-free f/m: [0.32605991 0.38440076 0.39150021 0.34867645]

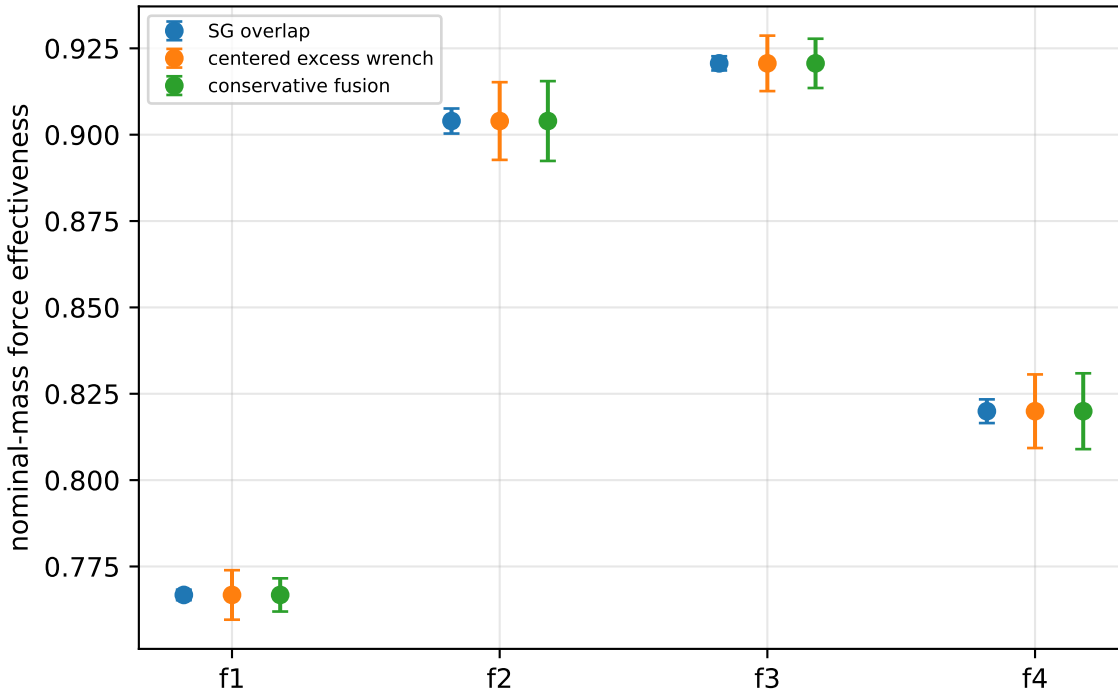
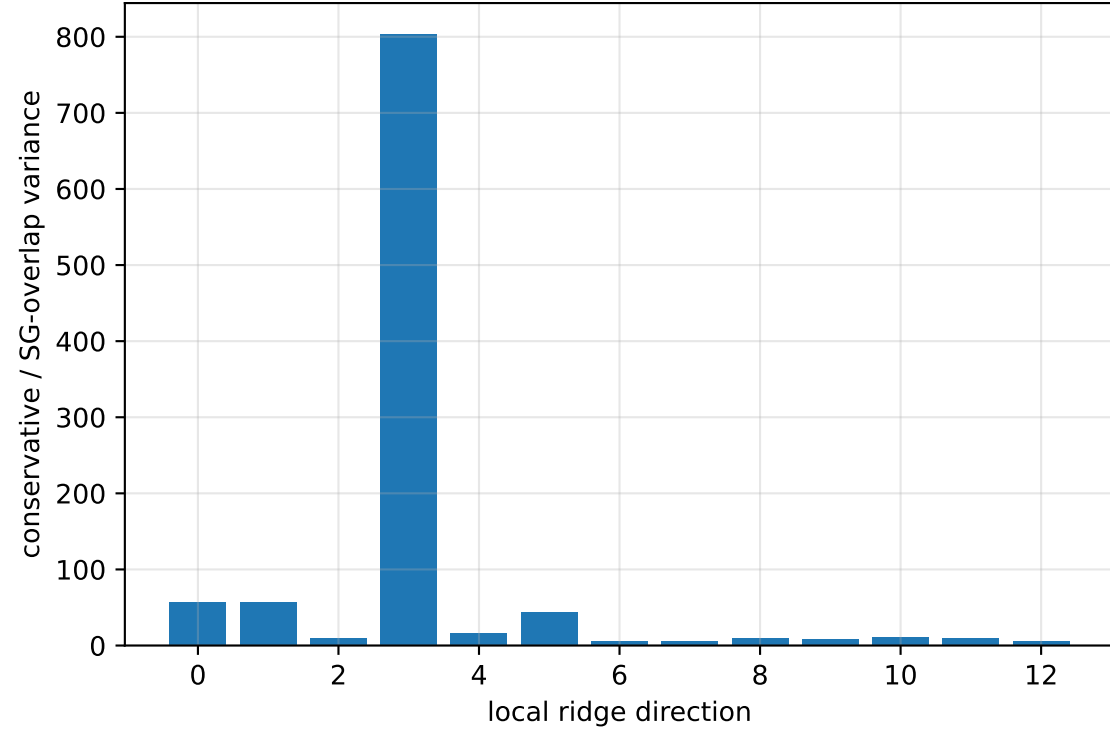
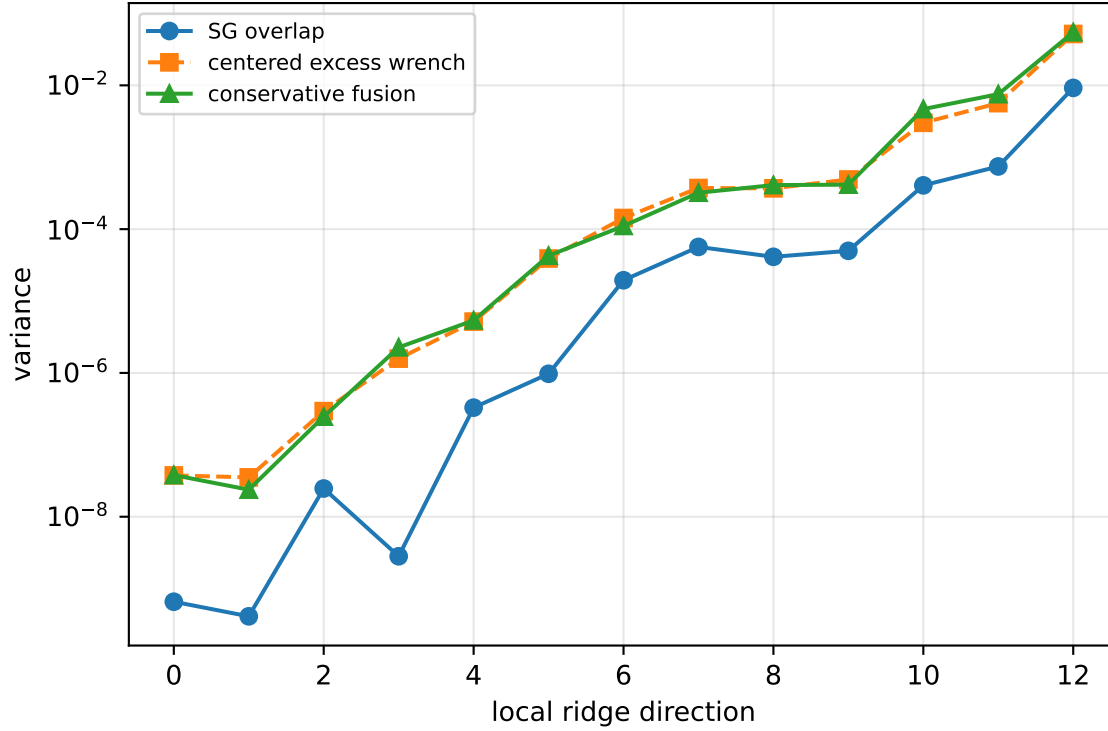
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

inertia_over_mass_zz_nominal: ridge spectrum (rank 13, nullity 1, ||v||=1.06e-13)



inertia_over_mass_zz_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 41.36$
wrench-to-acceleration recovery max error = 3.5×10^{-15}
exact scale gauge ridge index = 13 (alignment 1)

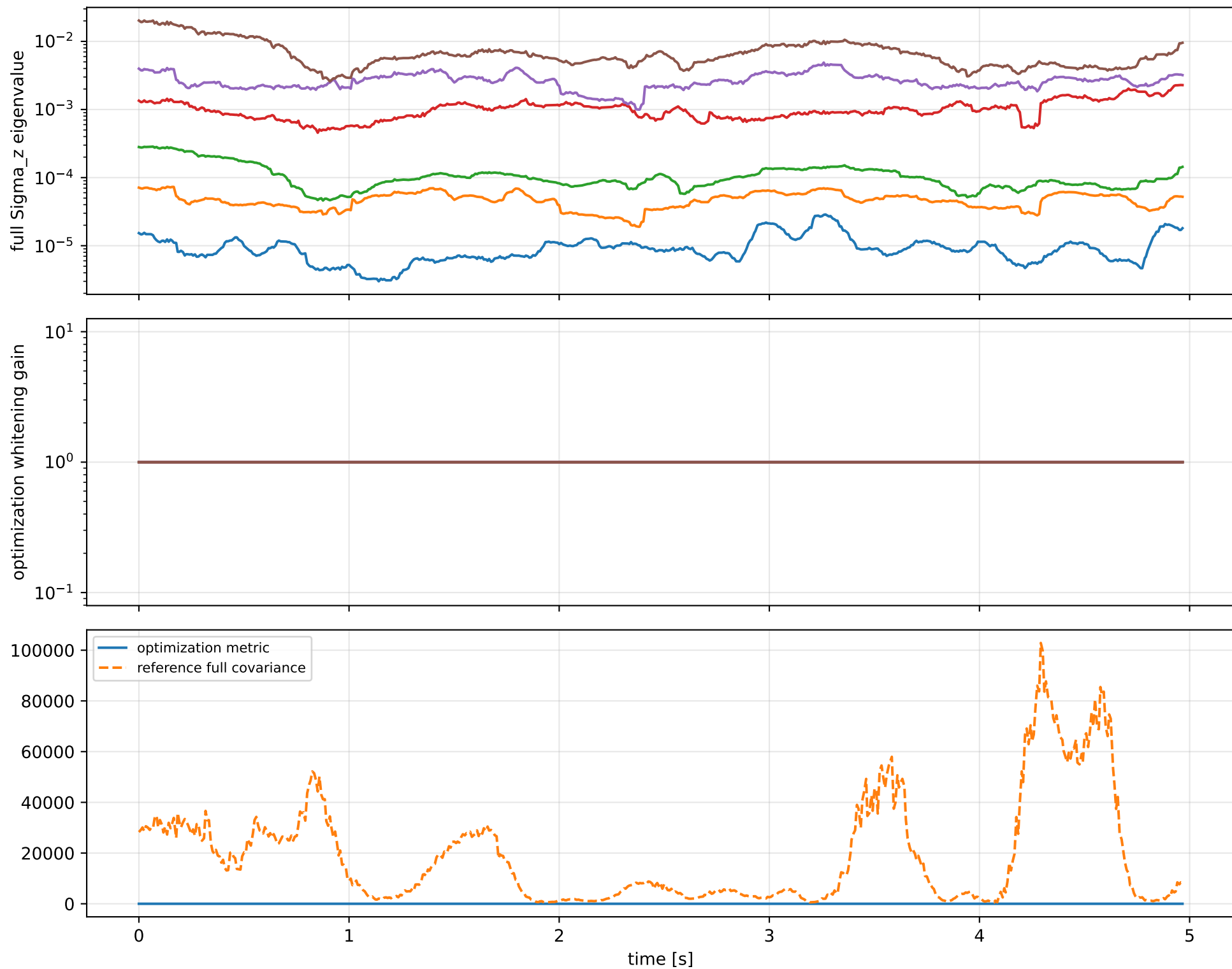
index 12: variance=0.05473, ratio=5.939
 $\log_{\text{mass}} = -0.0669$; second_moment_diag_1=+0.901
second_moment_diag_2=-0.236; second_moment_diag_3=-0.332
second_moment_offdiag_12=+0.00177; second_moment_offdiag_13=-0.00352
second_moment_offdiag_23=+0.006; cog_x=-0.000595
cog_y=-0.000996; cog_z=+0.000768
 $\log_{\text{force_eff_1}} = -0.0631$; $\log_{\text{force_eff_2}} = -0.0622$
 $\log_{\text{force_eff_3}} = -0.0766$; $\log_{\text{force_eff_4}} = -0.0646$

index 11: variance=0.007536, ratio=10.12
 $\log_{\text{mass}} = -0.152$; second_moment_diag_1=+0.0954
second_moment_diag_2=+0.893; second_moment_diag_3=-0.225
second_moment_offdiag_12=-0.0744; second_moment_offdiag_13=+0.128
second_moment_offdiag_23=+0.0433; cog_x=+0.00474
cog_y=+0.000943; cog_z=+0.000651
 $\log_{\text{force_eff_1}} = -0.175$; $\log_{\text{force_eff_2}} = -0.14$
 $\log_{\text{force_eff_3}} = -0.123$; $\log_{\text{force_eff_4}} = -0.173$

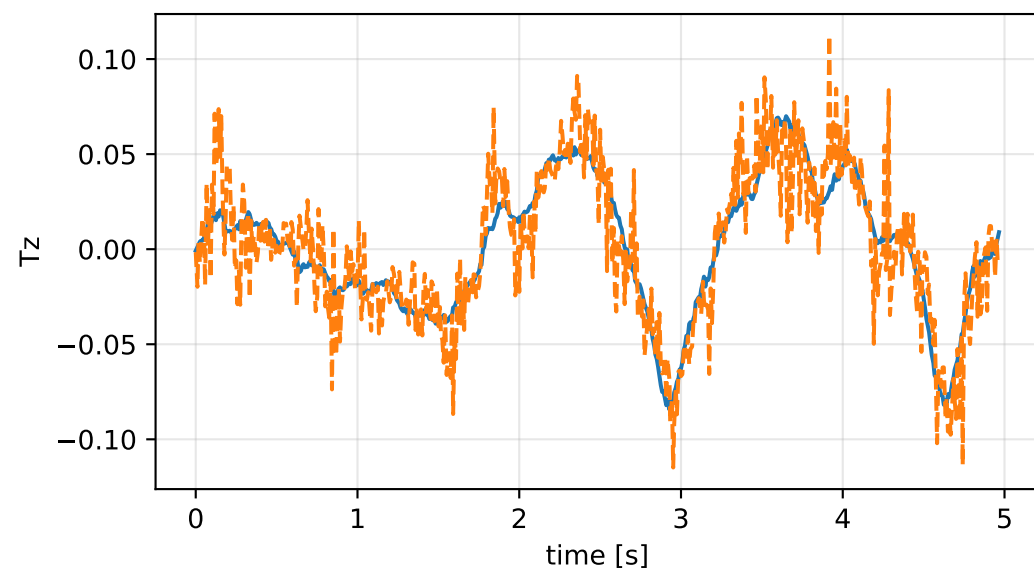
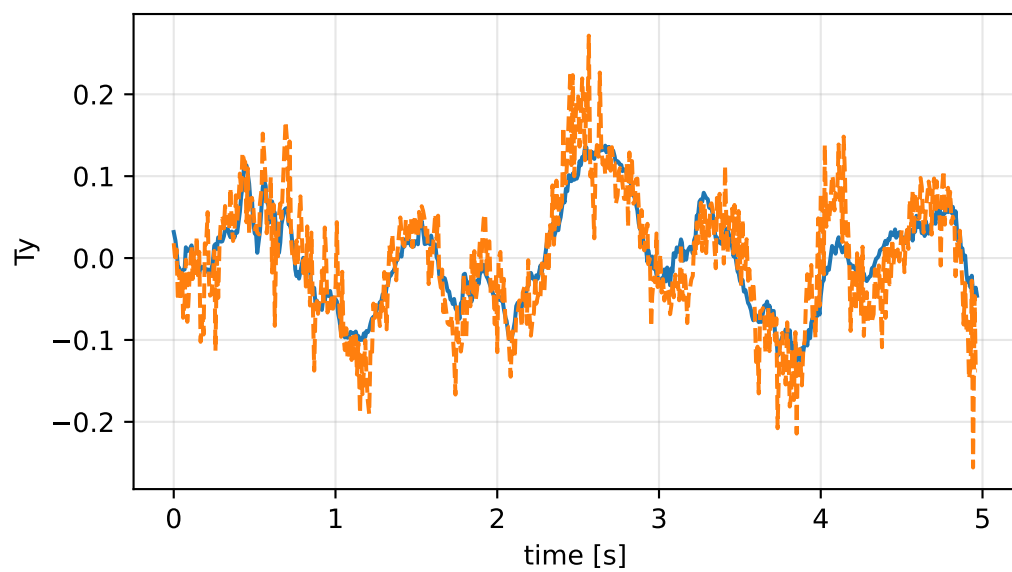
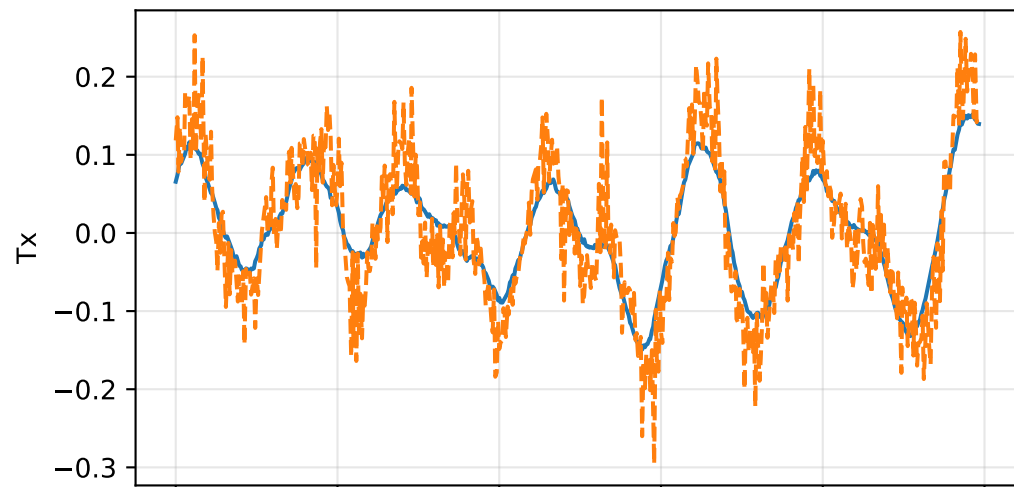
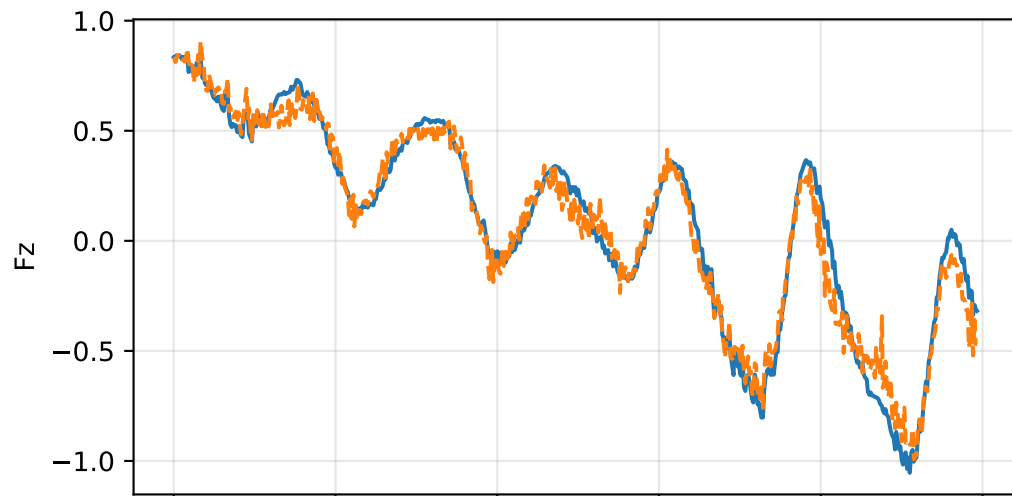
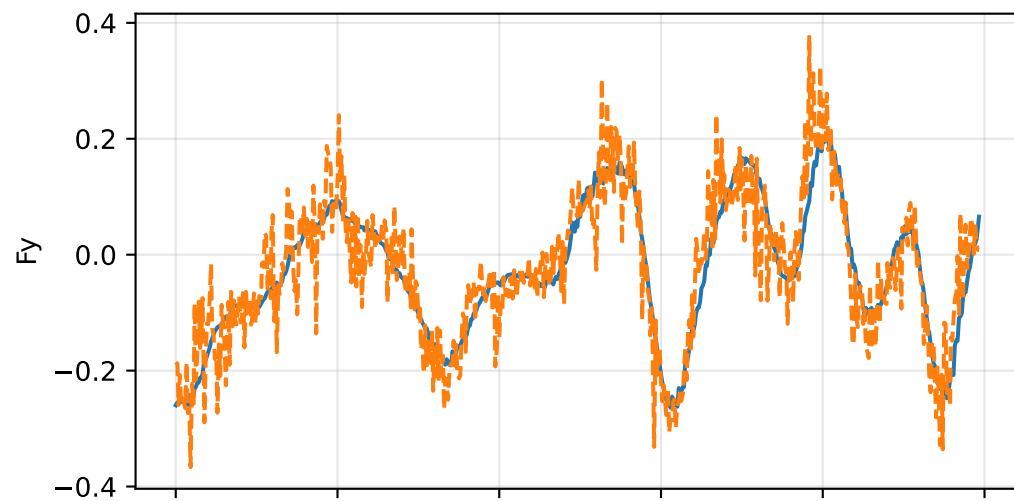
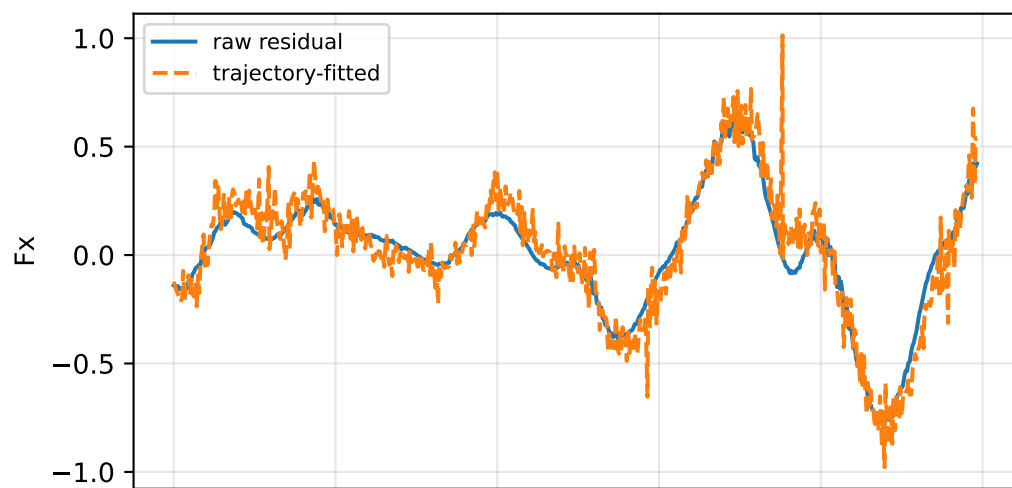
index 10: variance=0.004675, ratio=11.51
 $\log_{\text{mass}} = +0.0249$; second_moment_diag_1=-0.0108
second_moment_diag_2=-0.138; second_moment_diag_3=+0.0332
second_moment_offdiag_12=-0.491; second_moment_offdiag_13=+0.854
second_moment_offdiag_23=+0.0163; cog_x=+0.00769
cog_y=+0.00238; cog_z=-0.000531
 $\log_{\text{force_eff_1}} = -0.0321$; $\log_{\text{force_eff_2}} = +0.0563$
 $\log_{\text{force_eff_3}} = +0.059$; $\log_{\text{force_eff_4}} = +0.00772$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_zz_nominal: optimization vs reference covariance



inertia_over_mass_zz_nominal: wrench / closure (force max $4.55\text{e-}15$, torque max $2.15\text{e-}16$)



inertia_over_mass_zz_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_zz_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: 0aabc66d65b6ca544bf367511021d32b9f332e43a03fe96c3ce28f09a6e7e577

data objective: 316.005610027

prior objective: 5.94622588517e-07

total objective: 316.005610621

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_zz_nominal (inertia_over_mass_m2)

components: zz

target: [0.05485297]

std: [1.e-06]

final: [0.05485297]

error: [1.09052518e-09]

standardized residual: [0.00109053]

factor objective: 5.94622588517e-07